

National Health and Nutrition Examination Survey

2017-2018 Data Documentation, Codebook, and Frequencies

Body Measures (BMX_J)

Data File: BMX_J.xpt

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Last Revised: NA

Component Description

NHANES body measures data are used to monitor trends in infant and child growth, to estimate the prevalence of overweight and obesity in U.S. children, adolescents, and adults, and to examine the associations between body weight and the health and nutritional status of the U.S. population.

The measurements and target age groups for the NHANES 2017–2018 body measures component are as follows:

- Weight: All ages
- Head circumference: birth through 6 months of age
- Recumbent length: birth through 47 months of age
- Standing height: 2 years and older
- Upper leg length: 8 years and older
- Upper arm length: 2 months of age and older
- Mid-upper arm circumference: 2 months of age and older
- Waist circumference: 2 years of age and older
- Hip circumference: 12 years of age and older

Eligible Sample

All survey participants were eligible for the body measures component. There were no medical, safety, or other exclusions for the body measurements protocol. The health technicians used their discretion to obtain as many measures as practical for persons who used a wheelchair.

Protocol and Procedure

The body measures data were collected, in the Mobile Examination Center (MEC), by trained health technicians. The health technician was assisted by a recorder during the body measures examination. The participant's age at the time of the screening interview determined the body measures examination protocol. In some instances, the age at the screening interview and age at the time of the health examination differed by several weeks. The Demographics data file includes variables for age in years at screening (RIDAGEYR) for all participants and age in months at examination (RIDEXAGM) for participants who were less than 240 months of age at the time of examination.

Arm and leg measurements were made on the right side of the body. If a participant had an amputation, medical condition, or medical appliance, such as a cast, that prevented measurements from being taken on the right side of the body, the health technician took measurements on the left side. The body measurements file does not identify participants who had amputations because that information may be considered identifiable and pose a

disclosure concern. The body weight data for participants who had limb amputations were set to "missing".

This data file includes body measures for women who were pregnant at the time of their health examination. Pregnancy status at the time of the health examination is indicated by the variable, RIDEXPRG, in the Demographic data file. RIDEXPRG values are reported for women 20–44 years of age. RIDEXPRG for several pregnant women who were outside of this age range are not reported due to disclosure concerns. The body measures data for these participants are not reported.

Please refer to [the NHANES 2017–2018 Anthropometry Procedures Manual](#) for further details on obtaining body measurements.

Quality Assurance & Quality Control

The NHANES health technicians completed a 2-day training program with survey staff and an expert anthropometrist. The training included an overview of the component, general guidelines and technical skills for each measurement, and demonstrations conducted by the expert examiner with volunteer subjects. The expert examiner reviewed and demonstrated the proper technique to use for each measurement. Supervised practice exercises followed, conducted with several volunteer subjects, including infants, children, and adults. The chief health technician, at each of the MECs, monitored staff performance in the field. Health technician performance was also monitored using direct observation, data reviews, and periodic expert examiner (gold standard comparison) evaluations.

The body measures examination rooms in each of the MECs were identical with respect to layout and equipment. Scheduled equipment calibration was performed by the health technicians and verified by supervisory staff. The Anthropometry Procedures Manual includes detailed descriptions of the quality assurance and quality control measures that are used in the NHANES anthropometry/body measures component.

Data Processing and Editing

The 2017–2018 data were reviewed for unusual and erroneous values. Review criteria were based on the NHANES 1999–2016 body measurement data. During the data review, values that were above the 99th percentile or below the 1st percentile, for a particular age or age-gender group, were flagged for review. When records were flagged, the entire body measurements record was reviewed for reasonableness. Subject characteristics, such as height, weight, age, and gender were taken into consideration. Values that were determined to be unrealistic were deleted from the file. None of the original body measures data were changed and there are no imputed values in this file.

Body Mass Index (BMXBMI):

Body Mass Index (BMI) was calculated as weight in kilograms divided by height in meters squared, and then rounded to one decimal place.

Analytic Notes

Component status code: A final body measures component status code (BMDSTATS) provides analysts with a quick method of identifying survey participants with complete or partial body measurement data.

Unusual values: Unusual body measures values were noted during the review of the data. Typically, unusual values occurred when a subject was extremely short, tall, overweight, or underweight. In addition, the upper arm length (BMXARML) and upper leg length (BMXLEG) values may be affected by extreme amounts of adipose tissue. Analysts should examine the

distributions of the body measurements carefully and consider whether or not it is appropriate to include or exclude extreme values in a given analysis.

Comment codes: Comment codes were added by the health technicians, during data collection, to document problems or situations that arose during the body measures examination. For example, the variable BMIWT is a comment code for the body weight measurement. If a participant did not change into the NHANES exam gown, a code of BMIWT of "3" denoting "clothing worn" was made in the record. Analysts should review the comment code information for each of the body measures prior to data analysis.

Sample weights: The NHANES examination sample weights should be used to analyze the body measurement data. Please refer to the [NHANES Analytic Guidelines](#) and the online [NHANES Tutorial](#) for further details on the use of sample weights and other analytic issues.

References

- Centers for Disease Control and Prevention, National Center for Health Statistics, [NHANES III Anthropometric Procedures Video](#)
- Lohman TG, Roche AF, Martorell R, editors. Anthropometric Standardization Reference Manual. Abridged ed. Champaign, IL: Human Kinetics Books; 1988.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 0 YEARS - 150 YEARS

BMDSTATS - Body Measures Component Status Code

Variable Name: BMDSTATS
SAS Label: Body Measures Component Status Code
English Text: Body Measures Component status Code
Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Complete data for age group	8019	8019	
2	Partial: Only height and weight obtained	295	8314	
3	Other partial exam	283	8597	
4	No body measures exam data	107	8704	
.	Missing	0	8704	

BMXWT - Weight (kg)

Variable Name: BMXWT

SAS Label: Weight (kg)

English Text: Weight (kg)

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
3.2 to 242.6	Range of Values	8580	8580	
.	Missing	124	8704	

BMIWT - Weight Comment

Variable Name: BMIWT

SAS Label: Weight Comment

English Text: Weight Comment

Target: Both males and females 0 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Could not obtain	11	11	
3	Clothing	382	393	
4	Medical appliance	23	416	
.	Missing	8288	8704	

BMXRECUM - Recumbent Length (cm)

Variable Name: BMXRECUM

SAS Label: Recumbent Length (cm)

English Text: Recumbent Length (cm)

Target: Both males and females 0 YEARS - 47 MONTHS

Code or Value	Value Description	Count	Cumulative	Skip to Item
49.1 to 111.3	Range of Values	894	894	
.	Missing	7810	8704	

BMIRECUM - Recumbent Length Comment

Variable Name: BMIRECUM

SAS Label: Recumbent Length Comment

English Text: Recumbent Length Comment

Target: Both males and females 0 MONTHS - 47 MONTHS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Could not obtain	24	24	
3	Not straight	0	24	
.	Missing	8680	8704	

BMXHEAD - Head Circumference (cm)

Variable Name: BMXHEAD
SAS Label: Head Circumference (cm)
English Text: Head Circumference (cm)
Target: Both males and females 0 YEARS - 6 MONTHS

Code or Value	Value Description	Count	Cumulative	Skip to Item
32.4 to 47.9	Range of Values	194	194	
.	Missing	8510	8704	

BMIHEAD - Head Circumference Comment

Variable Name: BMIHEAD

SAS Label: Head Circumference Comment

English Text: Head Circumference Comment

Target: Both males and females 0 MONTHS - 6 MONTHS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Could not obtain	0	0	
.	Missing	8704	8704	

BMXHT - Standing Height (cm)

Variable Name: BMXHT

SAS Label: Standing Height (cm)

English Text: Standing Height (cm)

Target: Both males and females 2 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
78.3 to 197.7	Range of Values	8016	8016	
.	Missing	688	8704	

BMIHT - Standing Height Comment

Variable Name: BMIHT

SAS Label: Standing Height Comment

English Text: Standing Height Comment

Target: Both males and females 2 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Could not obtain	32	32	
3	Not straight	67	99	
.	Missing	8605	8704	

BMXBMI - Body Mass Index (kg/m2)****Variable Name:** BMXBMI**SAS Label:** Body Mass Index (kg/m**2)**English Text:** Body Mass Index (kg/m**2)**Target:** Both males and females 2 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
12.3 to 86.2	Range of Values	8005	8005	
.	Missing	699	8704	

BMXLEG - Upper Leg Length (cm)

Variable Name: BMXLEG

SAS Label: Upper Leg Length (cm)

English Text: Upper Leg Length (cm)

Target: Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
24.8 to 55	Range of Values	6703	6703	
.	Missing	2001	8704	

BMILEG - Upper Leg Length Comment

Variable Name: BMILEG

SAS Label: Upper Leg Length Comment

English Text: Upper Leg Length Comment

Target: Both males and females 8 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Could not obtain	334	334	
.	Missing	8370	8704	

BMXARML - Upper Arm Length (cm)

Variable Name: BMXARML
SAS Label: Upper Arm Length (cm)
English Text: Upper Arm Length (cm)
Target: Both males and females 2 MONTHS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
9.4 to 49.9	Range of Values	8177	8177	
.	Missing	527	8704	

BMIARML - Upper Arm Length Comment

Variable Name: BMIARML

SAS Label: Upper Arm Length Comment

English Text: Upper Arm Length Comment

Target: Both males and females 2 MONTHS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Could not obtain	347	347	
.	Missing	8357	8704	

BMXARMC - Arm Circumference (cm)

Variable Name: BMXARMC
SAS Label: Arm Circumference (cm)
English Text: Arm Circumference (cm)
Target: Both males and females 2 MONTHS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
11.2 to 56.3	Range of Values	8173	8173	
.	Missing	531	8704	

BMIARMC - Arm Circumference Comment

Variable Name: BMIARMC
SAS Label: Arm Circumference Comment
English Text: Arm Circumference Comment
Target: Both males and females 2 MONTHS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Could not obtain	350	350	
.	Missing	8354	8704	

BMXWAIST - Waist Circumference (cm)

Variable Name: BMXWAIST

SAS Label: Waist Circumference (cm)

English Text: Waist Circumference (cm)

Target: Both males and females 2 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
40 to 169.5	Range of Values	7601	7601	
.	Missing	1103	8704	

BMIWAIST - Waist Circumference Comment

Variable Name: BMIWAIST
SAS Label: Waist Circumference Comment
English Text: Waist Circumference Comment
Target: Both males and females 2 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Could not obtain	437	437	
.	Missing	8267	8704	

BMXHIP - Hip Circumference (cm)

Variable Name: BMXHIP

SAS Label: Hip Circumference (cm)

English Text: Hip Circumference (cm)

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
66.6 to 179	Range of Values	6039	6039	
.	Missing	2665	8704	

BMIHIP - Hip Circumference Comment

Variable Name: BMIHIP

SAS Label: Hip Circumference Comment

English Text: Hip Circumference Comment

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Could not obtain	270	270	
.	Missing	8434	8704	